

Association Transformations in Multi-Tier Structures

1. Introduction

This paper presents a method for learning transformations over multi-tier representations, with a particular focus on the modification of cross-tier relations (i.e., associations between tiers), which is widely used in linguistics. We consider a specific but linguistically important type of transformations where the nodes remain unchanged from input to output, while only the association relation is modified subject to non-crossing constraints. We formalize these representations as temporally-ordered tiers with a binary association relation, and the learning problem is therefore framed as inferring the which association relations are preserved, deleted, or newly introduced from observed input-output pairs.

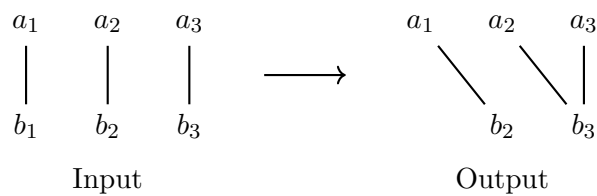


Figure 1: Learning association transformations between tiers.

Such multi-tiered structures have been widely used in linguistic theory, such as tone-syllable associations in autosegmental phonology (Goldsmith, 1976) prosodic or syllable structure (Hayes, 1980; Sagey, 1986). In such representations, linguistic information of different aspects are organized across multiple temporally ordered tiers. Associations between tiers encode which elements are linked at a given point in time, thereby capturing relationships across different dimensions of the structure. We refer this type of graph as *autosegmental graphs* (ASGs). More generally, such graphs can be viewed as bipartite graphs, where nodes with shared properties are organized into the same tiers. Since each tier is linearly ordered, associations are subject to non-crossing constraint (NCC, Bird and Ellison, 1994; Goldsmith, 1976). Previous work has modeled these structures using finite-state approaches (Wiebe, 1992; Bird and Ellison, 1994; Yli-Jyrä, 2013, 2015). These approaches typically convert structured representations into strings or synchronized tapes to leverage finite-state machinery, or rely on hard-coded mechanisms to represent different types of associations.

In contrast, we propose an approach that operates directly over autosegmental graphs under a model-theoretic perspective. We model transformations as logical transductions that map input to output by updating the association relation based on local structural context (graph neighborhood) in the input. By checking graph containment relation, we search the context that an association change. We show that, under locality constraints, this class of transformations falls within the subregular hierarchy and can be characterized as

strictly local functions over structured representations. Furthermore, transformations that appear non-deterministic when modeled over strings can be expressed deterministically in this relational framework. To our knowledge, the problem of learning local transformations of association relations in multi-tier structures has not been previously addressed in this form.

2. Preliminaries

2.1. Autosegmental Graphs

Let Σ be a finite alphabet and Σ^* the set of all strings over Σ . For a string $w \in \Sigma^*$, let $|w|$ denote its length. Model-theoretically, a string can be defined as relational structures of the form $\langle D, \mathcal{R} \rangle$, where D is a finite domain of positions, and $\mathcal{R}$ is a set of relations (known as the model *signature*). We illustrate the framework using linguistic examples, where graphs are used to represent structural associations between tones, segments, syllables, and other prosodic units.

Definition 1 (Autosegmental Graph) *An autosegmental graph (ASG) is a labeled bipartite graph with multiple linearly-ordered tiers. Equivalently, it can be defined model-theoretically as a relational structure $\mathcal{A} = \langle D, \mathcal{T}, \alpha, \triangleleft \rangle$ where D is a finite domain of elements; each tier $T_i \subseteq D$ is a set of nodes $\mathcal{T} = \{T_1, \dots, T_n\}$ is a collection of tiers such that $T_1 \cup T_2 \dots \cup T_n = D$ and $T_i \cap T_j = \emptyset$ for all $i \neq j$. $\triangleleft$ is a binary successor relation defined between nodes on the same tier; $\alpha \subseteq T \times V$ is a set of non-crossing association relations across tiers.*

ASGs represent the idea that different aspects of linguistic structure are organized on distinct tiers while progressing simultaneously over time. Each tier encodes a particular type of information, such as tone (pitch information), segmental (tongue shape), syllabic (metrical) structure. Association relations link elements across tiers that are realized at the same temporal position.

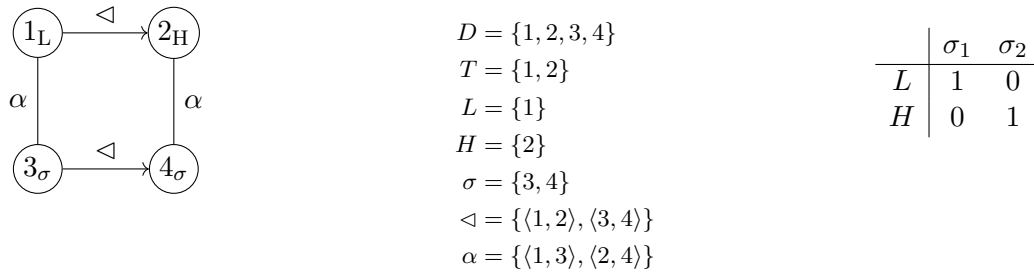


Figure 2: Three representations of a two-tier autosegmental graph.

Definition 2 (No Crossing Constraint (NCC)) *An ASG needs to obey No Crossing Constraint. Formally, For all $(x, y), (x', y') \in \alpha$, $x \triangleleft x' \Rightarrow (y \triangleleft y') \vee (y = y')$. and vice versa.*

Equivalently, we can represent an ASG using adjacency matrices, which encode the presence or absence of edges between nodes.

Definition 3 (Adjacency matrix) Given an ASG $\mathcal{A} = \langle D, T, V, \alpha, \triangleleft \rangle$, where T and V are two tiers, the association relation α can be represented as a binary matrix where $M_\alpha[i, j] = 1$ if $(x_i, y_j) \in \alpha$ where $x_i \in T, y_j \in V$; otherwise 0.

Definition 4 Let matrix A be an $m \times n$ matrix. Matrix B is a submatrix of A by taking k ($1 \leq k \leq m$) contiguous rows and v ($1 \leq v \leq n$) contiguous columns from A .¹

2.2. Logical Transduction

Using model theory (Strother-Garcia, 2019; Yolyan, 2025; Jardine, 2017) we define a multi-tiered structure as a model $\mathcal{M} := \langle D, \mathcal{R} \rangle$, where D is a finite domain of positions, and $\mathcal{R}$ is a set of relations (also known as the model *signature*). An autosegmental graph is defined over a signature consisting of unary predicates, which indicate tonal or segmental properties, such as H, L, V , and C , and binary relations that encode structural connections. In particular, we assume two binary relations: $\triangleleft$ and α . The relation $\triangleleft$ encodes the successor relation: $\triangleleft(x, y)$ indicates that x immediately follows y . For example, vowel hiatus can be defined as $\text{hiatus}(x, y) := V(x) \wedge V(y) \wedge \triangleleft(x, y)$. The relation $\alpha(x, y)$ encodes association, indicating that tone x is linked to segment y . These predicates, defined in model-theoretic terms, express exactly the same core ideas as autosegmental phonology (?), and reveal greater simplicity as we delve deeper into tonal processes.

Building on this view, we define a phonological transformation from an underlying representation (UR) to a surface representation (SR) is modeled as a relation between input and output structures (see more in ?). The signature of the output can be obtained by referring to the input signature. Formally, for any predicate $R(x)$ in the input signature, a faithful output predicate R' is defined as $R'(x) := R(x)$, meaning that an element x in the output domain has property R iff it has property R in the input.

Our proposal is that vowel-deletion-induced tonal processes preserve all structural relations in the AR except for α , which is subject to language-specific *faithfulness*, *deassociation*, and *reassociation*: $\alpha'(x, y) := (\alpha(x, y) \wedge \neg \text{deassoc}(x, y)) \vee \text{reassoc}(x, y)$. The deleted units remains in the domain but are not realized at the surface, since a licensing function ϕ_{license} restricts realization to associated units.

Table ?? lists all transductions from input to output. The licensing function ensures that only associated (i.e., fully specified) units surface in the SR. All unary relations are preserved from UR to SR, as is the successor relation $\triangleleft$.

The output association relation α' is defined as follows: an association line holds in the output if (i) it is present in the input and not removed by **deassoc**, or (ii) it is introduced by **reassoc**. Concretely, **deassoc** removes the association between the deleted vowel and its tone.

2.3. Association Learning

We now present an algorithm for learning local transformations of association relations from input-output pairs:

1. The term “submatrix” is generally defined by deleting arbitrary rows and/or columns from the original matrix. However, in our case, we require a more restricted definition: the remaining rows and columns must be *contiguous* in the original matrix, which reflects tonal locality.

Input: Input-output model-theoretic pairs $D = \{(I_1, O_1), (I_2, O_2), \dots, (I_n, O_n)\}$

Output: A grammar G of conjunctive negative constraints

$G, \alpha^-, \alpha^+, \alpha^\pm \leftarrow \emptyset$

$Q \leftarrow \{\alpha(x, y)\}$

$P \leftarrow \emptyset;$ // Rejected contexts

foreach $(I, O) \in D$ **do**

$\alpha^- \leftarrow \alpha^- \cup (\alpha_I \setminus \alpha_O) ;$ // Deleted sites

$\alpha^+ \leftarrow \alpha^+ \cup (\alpha_O \setminus \alpha_I) ;$ // Added sites

$\alpha^\pm \leftarrow \alpha^\pm \cup (\alpha_I \cap \alpha_O) ;$ // Faithful sites

end

while $Q \neq \emptyset$ **do**

$c \leftarrow Q.\text{Dequeue}()$

if $\forall g \in \alpha^*, c \sqsubseteq g$ **then**

if $\forall g \in \alpha^\pm, c \not\sqsubseteq g$ **then**

$G \leftarrow G \cup \{c\}$

end

else

foreach $s \in \text{NEXTSUPFACT}(c)$ **do**

if $s \not\sqsubseteq p$ for all $p \in P$ **then**

$Q.\text{Enqueue}(s)$

end

end

end

else

$P \leftarrow P \cup \{c\}$

end

end

return G

3. Case Study

3.1. Contour Formation

We begin with an idealized assumption: the tonal process involves no change to the segmental string, and the only modifications between the underlying representation (UR) and the surface representation (SR) are changes in association lines, namely DEASSOCIATION and REASSOCIATION. Although many phonological processes also involve segmental deletion, insertion, or feature changes, we restrict the current discussion to association-line modifications in order to isolate the learning problem.

The first step is to identify exactly what structural modification has taken place between the UR and the SR. Since the segmental and tonal tiers remain constant, the comparison is made directly over the association relation matrix.

For each cell in the association matrix:

That is, if an association present in the UR disappears in the SR, the operation is classified as deassociation; if a new association appears in the SR that was absent in the UR, the operation is classified as reassociation.

In Estoko, both operations are attested: one association line is removed while another is added. Therefore, the learner must infer both the deassociation environment and the reassociation environment separately. (maybe not maybe they can start together)

Hypothesis space for deassociation. Given a set of observed deassociation sites, the learner constructs candidate hypotheses describing the structural environments in which an association changes from 1 to 0. Each hypothesis corresponds to a local structural condition under which $\alpha(x, y)$ is removed.

The search proceeds from simple to more complex descriptions. At each step, the learner considers increasingly rich local configurations (factors) until a hypothesis is found that captures all and only the observed deassociation sites.

The most general hypothesis removes all associations:

$$\text{Hypothesis 1: } 1 \rightarrow 0$$

This hypothesis is immediately rejected if there exist associations in the data that remain unchanged.

The learner then considers one-factor hypotheses, in which deassociation is triggered by a single structural property of either tier:

$$\text{DEASSOC}(x, y) := \alpha(x, y) \wedge H(x)$$

$$\text{DEASSOC}(x, y) := \alpha(x, y) \wedge V(y)$$

These hypotheses predict that all associations involving a particular tone or segment are removed. They are rejected if counterexamples exist in which such associations are preserved.

Next, the learner considers two-factor hypotheses that combine information from both tiers:

$$\text{DEASSOC}(x, y) := \alpha(x, y) \wedge L(x) \wedge V(y)$$

This hypothesis restricts deassociation to associations between a specific tone and segment type.

Additional hypotheses incorporate local structural context, such as adjacent elements on a tier. For example, conditions may refer to neighboring segments or tones within a bounded window:

$$\text{DEASSOC}(x, y) := \alpha(x, y) \wedge L(x) \wedge \text{Next}_V(y, y')$$

where Next_V denotes adjacency on the segment tier.

More complex hypotheses combine multiple local conditions. For example, a three-factor hypothesis may refer to both the tone and the local segmental configuration:

$$\text{DEASSOC}(x, y) := \alpha(x, y) \wedge L(x) \wedge \text{Next}_V(y, y') \wedge V(y')$$

In general, the learner searches over hypotheses defined by local factors of increasing size, corresponding to progressively larger neighborhoods in the underlying graph. The goal is to identify the smallest hypothesis that correctly predicts all and only the observed deassociation sites.

4. Conclusion

This paper has argued that vowel-deletion-induced tonal processes can be uniformly captured as logical transductions over autosegmental representations (ARs). The observed typological variation follows from modifying a single relation, α , through the interaction of `deassoc` and `reassoc`.

A limitation of the current analysis is that the transduction relies on quantification to capture across-tier dependencies. An important direction for future work is to determine whether these processes can be simplified in a quantifier-free (QF) process. This work also extends naturally to learning over ARs. Recent work by Yolyan (2025) demonstrates that logical transductions are learnable by using a bottom-up factorial inference algorithms (Chandlee et al., 2019). This opens up more discussions about how linguistically meaningful representations can facilitate grammar inference.

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